Check for updates

FEBS Letters

FEBS PRESS science publishing by scientists

REVIEW ARTICLE

DNA-encoded chemical libraries – achievements and remaining challenges

Nicholas Favalli†, Gabriele Bassi†, Jörg Scheuermann and Dario Neri

Department of Chemistry and Applied Biosciences, Swiss Federal Institute of Technology (ETH Zürich), Zürich, Switzerland

Correspondence
J. Scheuermann and D. Neri, Department of Chemistry and Applied Biosciences, Swiss Federal Institute of Technology (ETH Zürich), Vladimir-Prelog-Weg 4, CH-8093 Zürich, Switzerland
Fax: +41 44 633 13 58
Tel: +41 44 633 77 74 (J.S.);
+41 44 633 74 01 (D.N.)
E-mails: joerg.scheuermann@pharma.ethz.ch; dario.neri@pharma.ethz.ch

†These authors have contributed equally.

(Received 20 March 2018, revised 11 April 2018, accepted 13 April 2018, available online 10 May 2018)

doi:10.1002/1873-3468.13068

Edited by Wilhelm Just

DNA-encoded chemical libraries (DECLs) are collections of compounds, individually coupled to DNA tags serving as amplifiable identification barcodes. Since individual compounds can be identified by the associated DNA tag, they can be stored as a mixture, allowing the synthesis and screening of combinatorial libraries of unprecedented size, facilitated by the implementation of split-and-pool synthetic procedures or other experimental methodologies. In this review, we briefly present relevant concepts and technologies, which are required for the implementation and interpretation of screening procedures with DNA-encoded chemical libraries. Moreover, we illustrate some success stories, detailing how novel ligands were discovered from encoded libraries. Finally, we critically review what can realistically be achieved with the technology at the present time, highlighting challenges and opportunities for the future.

Keywords: combinatorial chemistry; DNA; DNA-encoded chemical libraries; drug discovery; high-throughput DNA sequencing

The discovery of small organic ligands, capable of selective binding to protein targets of interest, represents a central problem in Chemistry, Biology, and Pharmaceutical Sciences. Until recently, the main approach for the discovery of binding molecules ("hits"), that may subsequently be improved using Medicinal Chemistry techniques, has relied on the screening of large collections of compounds ("chemical libraries"), one molecule at a time. In general, the screening of chemical libraries is possible if a suitable bioassay is in place. For example, chemical compounds can be individually tested in microtiter plates using high-throughput screening (HTS) procedures for their ability to inhibit the biochemical activity of a target enzyme. This approach, which requires complex logistics, is typically limited to the screening of libraries, which in most cases are smaller than one million compounds. It has been estimated that the construction of a conventional library, containing 10^6 individual molecules in sufficient quantity and quality for pharmaceutical screening campaigns, may cost between 400 million and 2 billion U.S. dollars [1]. While certain academic efforts for the HTS of chemical libraries have been implemented, those activities can rarely compete with the ones performed by large pharmaceutical companies, in terms of hit productivity, as well as size and diversity. Importantly, conventional HTS approaches often fail to deliver ligands of sufficient affinity and specificity for pharmaceutical applications. This

Abbreviations
AGP, alpha-1-acid glycoprotein; BTE, binder trap enrichment; CuAAC, Cu-catalyzed alkyne-azide cycloaddition; DECLs, DNA-encoded chemical libraries; DPAL, DNA-programmed photoaffinity labeling; ELT, encoded library technology; ESAC, encoded self-assembled chemical; GSK, GlaxoSmithKline; HTS, high-throughput screening; LNA, locked-nucleic acids; PCR, polymerase chain reaction; sEH, soluble epoxide hydrolase.

underlines the need for improved library synthesis and screening technologies.

DNA-encoded chemical libraries are large collections of organic molecules, individually coupled to DNA fragments, serving as amplifiable identification barcodes [2]. As molecules are individually tagged with DNA barcodes, library members can be stored as a mixture and interrogated by affinity capture procedures. The use of DNA tags as barcodes allows the construction and screening of chemical libraries of unprecedented size, potentially exceeding one billion compounds. The identity and relative quantity of individual compounds can be determined by polymerase chain reaction (PCR) procedures, followed by high-throughput DNA sequencing. The libraries used for affinity-based screening can be very large, since even few copies of compounds (i.e., library members present at low concentration) can be reliably detected and quantified, thanks to the amplification power of PCR.

The advent of DNA-encoded chemical libraries was inspired by the successful introduction of technologies for the construction and screening of encoded combinatorial libraries of proteins and other biomacromolecules. For example, the display on filamentous phage of antibody libraries allows the facile identification of human monoclonal antibodies, specific to target proteins of interest [3–6]. Similarly, large libraries of oligonucleotides ("aptamers") can be interrogated by affinity capture procedures on target proteins of interest and "decoded" by DNA sequencing [7,8]. While DNA acts as a barcode that drives the synthesis of the corresponding biomacromolecule (e.g., DNA transcription and translation), in the case of chemical libraries the individual DNA fragments merely serve as identification barcodes, without a biosynthetic relation to the corresponding organic molecule.

The origin of DNA-encoded chemical library (DECL) technology dates back to the proposal of Brenner and Lerner to use beads for the alternate synthesis of polymers (e.g., peptides) and of cognate oligonucleotides serving as amplifiable barcodes [2,9]. In 2004, three groups independently described that DNA-encoded chemical libraries can be synthesized and screened without beads [10–12]. Since then, in virtually all cases, DNA-encoded chemical libraries are synthesized and screened by direct coupling of organic molecules to the corresponding DNA tags [1,13–21]. There are, however, multiple strategies for the synthesis of encoded libraries, as illustrated in a later section of this manuscript.

We normally distinguish between two main types of DNA-encoded chemical libraries: (i) "single-pharmacophore libraries", in which individual compounds (no matter how complex) are attached to one DNA fragment; and (ii) "dual-pharmacophore libraries", in which pairs of compounds are coupled to the extremity of the complementary strands of the DNA heteroduplex (Fig. 1). In this second example, binding fragments are identified, which need to be chemically linked at a later stage, in order to yield organic molecules which can be used in the absence of DNA. An appropriate linker optimization procedure may help further improve the affinity constants toward the protein target [22].

In full analogy to antibody phage display libraries or aptamer libraries, DNA-encoded chemical libraries

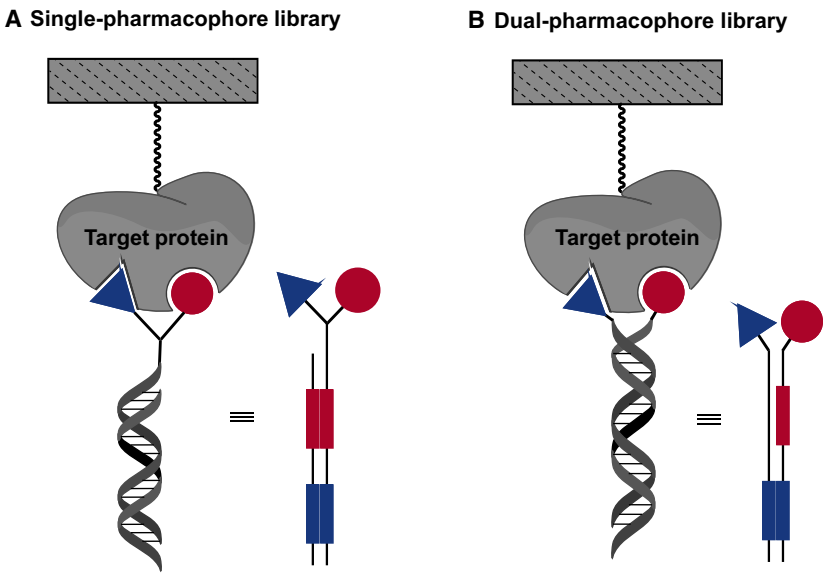


Fig. 1. Schematic representation of (A) a single-pharmacophore DNA-encoded chemical library and (B) of a dual-pharmacophore DNA-encoded chemical library. In the scheme, the building blocks (triangles and circles) and the corresponding DNA barcodes (rectangles) are depicted using the same color.

18733468, 2018, 12, Downloaded from https://febs.onlinelibrary.wiley.com/doi/10.1002/1873-3468.13068 by Schweizerische Akademie Der Medizinischen Wissenschaften (Samw), Wiley Online Library on [03/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

are typically screened by performing affinity capture procedures on a target protein of interest, immobilized on a solid support. For example, biotinylated proteins can be conveniently immobilized on streptavidin-coated magnetic beads [23] or on streptavidin-coated resin [17]. Molecules within the library, which are able to interact with the target protein, are preferentially enriched using this procedure, while nonbinding compounds can be washed away. A comparative analysis of the relative abundance of the DNA tags in the library, before and after affinity capture, provides direct information about the identity of molecules, which are enriched on the target protein of interest. In general, it is good practice to confirm the enrichment results by resynthesis of the hit compounds and by dedicated affinity measurements ("hit validation"). In addition to compound resynthesis, hit validation strategies performed "on DNA" can be considered, as the nucleic acid may facilitate resynthesis, MS detection, solubility and affinity measurements with certain experimental procedures [24,25].

The scope of DNA-encoded chemical library technology has been extended by exploiting the chemical opportunities offered by peptide nucleic acids (PNAs) [26]. The chemical structure of PNAs allows the execution of broader set of chemical reactions, which may not always be possible with DNA [27]. However, since PNA molecules cannot be amplified by PCR, PNA-encoded libraries need to be analyzed by hybridization with complementary DNA molecules [26–28] rather than by PCR amplification and direct DNA sequencing.

Encoding strategies

In theory, it would be conceivable to generate a DNA-encoded chemical library by coupling one by one individual compounds to distinctive oligonucleotides, serving as identification barcodes. Such an approach, however, would be expensive, unpractical, and inefficient for many reasons. Firstly, it would be difficult to find millions of organic molecules, suitable for coupling to DNA. Secondly, it would be very expensive to synthesize or purchase millions of oligonucleotides. Thirdly, the freedom to generate and encode novel molecular designs would be severely limited.

Luckily, efficient strategies for the synthesis and encoding of chemical libraries have been developed, which are modular in nature and which require a limited number of oligonucleotides and of chemical building blocks. It is convenient to make a distinction between single- and dual-pharmacophore chemical libraries.

Single-pharmacophore libraries

The most commonly used strategy for encoding single-pharmacophore libraries is called "DNA-recorded synthesis" and relies on the use of split-and-pool procedures (Fig. 2A,B) [29,30]. The basic concept behind this technology relates to the synthesis of chemical compounds using multiple steps, each of which is "recorded" by the addition of a DNA fragment that uniquely identifies a given chemical transformation. For illustrative purposes, we could consider the stepwise assembly of a library, generated by the reaction of two sets of building blocks (e.g., 1000 × 1000 building blocks, leading to 10^6 chemical entities). In this case, it is convenient to couple the first 1000 building blocks to 1000 different oligonucleotides, which unambiguously identify the attached cognate chemical moiety. In most cases, the oligonucleotides that are used are similar in sequence, but differ in a central region (serving as "barcode"). The 1000 encoded compounds can now be mixed, used as a pool and split into 1000 reaction vessels for a subsequent chemical transformation (e.g., with 1000 building blocks). At the end of the second set of reactions, a barcode could be added, providing an "identifier" for the chemical transformation which has been performed.

Fig. 2A depicts a strategy for the DNA-recorded synthesis of a single-pharmacophore library, consisting of three sets of building blocks. Initially, a first set of building blocks (shown in red) is coupled to amino-tagged oligonucleotides, featuring a distinctive central sequence as barcode. After the coupling step, the conjugates can be pooled and split into several vessels, for the reaction with a second set of building blocks (shown in blue). After the reaction step, oligonucleotides can be extended by the use of a splint ligation procedure [21,31], thus incorporating barcodes for the second reaction step (i.e., identifiers for the chemical nature of the individual building blocks used for the second reaction). After a further suitable pool-and-split procedure, a third set of chemical transformations can be performed (with building blocks depicted in green), thus generating the final compounds, consisting of three sets of building blocks. The chemical identity of the third reaction can be encoded by annealing with a set of partially complementary oligonucleotides, followed by a Klenow polymerization step [21,29,32,33]. This procedure is efficient as it makes use of one oligonucleotide for each building block used for library construction. For example, in the case of a library consisting of 1000 × 1000 × 1000 building blocks (10^9 compounds), only 3000 oligonucleotides and chemical building blocks will be needed.

18733468, 2018, 12, Downloaded from https://febs.onlinelibrary.wiley.com/doi/10.1002/1873-3468.13068 by Schweizerische Akademie Der Medizinischen Wissenschaften (Samw), Wiley Online Library on [03/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

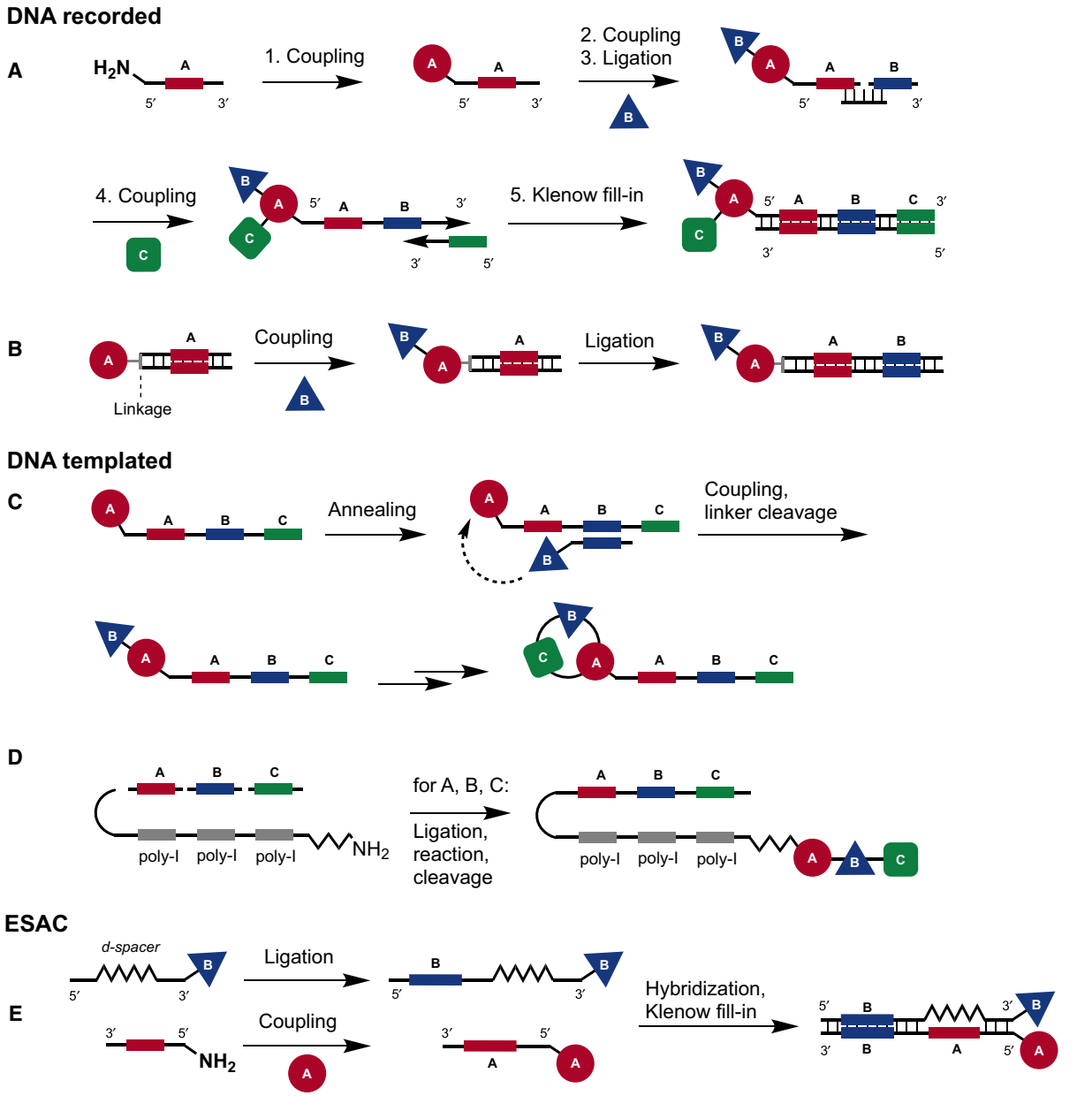


Fig. 2. (a) Schematic representation of strategies for DNA-recorded synthesis. In the first reaction cycle, chemical building blocks are encoded by direct coupling with 5′ amino-modified oligonucleotides containing individual coding DNA-sequences (Code "A"). After a split-and-pool step which introduces the second set of building blocks B the DNA-code "B" is introduced by enzymatic DNA-ligation. In the last reaction cycle, the introduced building can be encoded by annealing with a partially complementary oligonucleotide containing code "C", followed by the fill-in of the DNA heteroduplex by Klenow polymerase. (b) In a variation of the encoding procedures described in (a), the organic moiety is connected by a linker (termed "headpiece") to a double-strand DNA which is extended by subsequent ligation with coding DNA heteroduplexes, in parallel with the split-and-pool-based synthesis. (c) In DNA-templated synthesis, preformed DNA-template molecules containing coding parts are annealed with code-specific reagent oligonucleotides, which mediate the transfer of the chemical moiety by its high effective molarity. After cleavage of the chemical moiety from the reagent oligonucleotide, it can be removed and the template undergo a new round of template-based synthesis. (d) In a further implementation of this procedure, a template containing poly-inosine (poly-I) segments allows the annealing with various code-building block oligonucleotide conjugates. The building blocks are then transferred from the code-building block oligonucleotide conjugates to the main oligonucleotide template which, after hybridization, is encoded by ligation. (e) In the ESAC approach, two partially complementary sublibraries A and B are combinatorially assembled. A Klenow fill-in reaction facilitates the transfer of code B onto the complementary strand, bearing code A. Following target-based selection, PCR amplification and DNA sequencing allows the identification of the preferentially enriched pairs of building blocks.

18733468, 2018, 12, Downloaded from https://febs.onlinelibrary.wiley.com/doi/10.1002/1873-3468.13068 by Schweizerische Akademie Der Medizinischen Wissenschaften (Samw), Wiley Online Library on [03/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

An alternative procedure, which is frequently used in the industrial environment, makes use of double-stranded DNA fragments, connected by a chemically modifiable linker, serving as starting structure for the construction of encoded libraries [17] (Fig. 2B). Also in this case, sets of chemical transformations are performed using split-and-pool procedures, leading to large combinatorial diversity. As the nascent DNA structure is in double-stranded format, it is convenient to encode each chemical transformation by the ligation of an additional double-stranded DNA fragment. The procedure can be iterated. In this case, a library consisting of $1000 \times 1000 \times 1000$ building blocks will require 6000 oligonucleotides (i.e., 3000 double-stranded DNA fragments).

An alternative strategy for library synthesis and encoding, first developed by the group of David Liu [34,35], relies on the use of long presynthesized oligonucleotides [bearing suitable coding sequences (here depicted as A, B and C)], serving as "templates" for the stepwise assembly of nascent organic molecules (Fig. 2C). According to this strategy, oligonucleotides carrying suitable reactive building blocks are hybridized to the cognate template sequences, thus facilitating a subsequent reaction step with a nascent molecule attached to the template. After a cleavage and oligonucleotide removal step, additional sets of reactions can be performed, thus leading to the sequential assembly of complex chemical structures. The transfer of building blocks may be facilitated by the high effective molarity associated with heteroduplex formation [11]. DNA-templated strategies have been particularly useful for the synthesis of encoded polypeptidic and of macrocyclic libraries [36]. A variation of the scheme, termed "Yoctoliter reactor™" has been proposed by Vipergen, featuring templated reactions which occur on DNA three-way junctions [18].

In order to avoid encoding errors and side reactions, the DNA-templated synthesis requires a high-fidelity annealing between complementary oligonucleotide sequences (Fig. 2C). This feature may represent a potential limitation of the technology. In order to solve this problem, Li and coworkers [37] have proposed a modified template architecture, featuring a "universal template" that contains poly-inosine segments, serving as promiscuous hybridization stretches (Fig. 2D). Short oligonucleotide derivatives, carrying suitable reactive building blocks, can be sequentially hybridized and ligated to the universal template, upon formation of a hairpin structure. The strategy allows the stepwise synthesis of chemical structures, accompanied by the incorporation of barcodes which can be PCR amplified and which unambiguously identify the chemical nature of the building blocks for all library members.

Dual-pharmacophore chemical libraries

Dual-pharmacophore libraries are often constructed on the basis of a synthetic strategy, leading to "encoded self-assembled chemical (ESAC) libraries". This technology, initially proposed by our group in 2004 [10], makes use of two sets of partially complementary oligonucleotides, each containing a distinctive DNA barcode (Fig. 2E). The two sets of chemically modified oligonucleotides constitute "sublibraries", which can be mixed and hybridized to form stable DNA heteroduplexes, which display pairs of building blocks. For example, the combinatorial assembly of two sublibraries with 1000 chemical moieties each leads to a 1-million membered ESAC library.

In order to generate ESAC libraries with DNA structures, which allow the identification of the chemical nature of the pairs of building blocks by DNA sequencing, an encoding strategy featuring the use of oligonucleotides with abasic sites has been described [31]. One sublibrary (in this case, sublibrary A, Fig. 2E) is synthesized by coupling a set of building blocks to 5′-amino-modified oligonucleotides, which unambiguously code for the cognate compounds. A second set of building blocks is first coupled to a single oligonucleotide bearing a *d-spacer* region. The resulting conjugates can be subsequently encoded by splint ligation with distinct code-bearing oligonucleotides, yielding the sublibrary B (Fig. 2E). The *d-spacer* allows the hybridization of each member of sublibrary B with each member of the complementary sublibrary A, avoiding base pair mismatches and allowing the creation of very large combinatorial repertoires. A code-transfer strategy, which makes use of a Klenow fill-in DNA polymerization step, allows the transfer of code B onto the complementary strand (sublibrary A, Fig. 2E). This feature allows the simultaneous determination of the binding fragments by PCR amplification and high-throughput DNA sequencing [31].

Dual-display libraries feature a higher degree of flexibility in the arrangement of the chemical building blocks compared to single-pharmacophore DECLs and may more easily reach adjacent binding sites on the surface of the cognate protein target. On the other hand, as soon as a combination of building blocks (A and B) has been identified as preferential binding pair, various linkers need to be tested, in order to yield binding molecules devoid of DNA, which can be used for practical applications. For that purpose, the use of a set of predefined bifunctional scaffolds and the implementation of affinity measurements on locked

18733468, 2018, 12, Downloaded from https://febs.onlinelibrary.wiley.com/doi/10.1002/1873-3468.13068 by Schweizerische Akademie Der Medizinischen Wissenschaften (Samw), Wiley Online Library on [03/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

nucleic acids (LNA) may facilitate the identification of the best linker [22].

Selection strategies

Several strategies for affinity-based selections using DNA-encoded chemical libraries have been proposed and implemented in practice [19,38]. These methods can be grouped into two main categories: (i) solid-phase selections, in which the target proteins are anchored on a solid support, allowing the physical isolation of preferential binders; and (ii) solution-phase selections, which do not require the need of immobilized protein for library screening.

Solid-phase selections

Solid-phase affinity-based selections make use of target proteins, which are immobilized on a solid support and subsequently incubated with a DNA-encoded chemical library, allowing the physical separation of preferential binders from the other library members, which can be washed away. Convenient supports for library capture include magnetic beads [23,29,32] or resin-filled tips [17]. Washing conditions and the use of detergents may greatly impact on the selection results [23,32]. After affinity capture, the barcodes of preferential binders are PCR amplified and submitted to a high-throughput sequencing procedure. A comparison of Illumina and 454 sequencing technologies has been reported [39]. In some cases, it may be preferable to incubate the library with a suitably tagged protein (e.g., a biotinylated protein) in solution, with a subsequent capture step on an affinity support (e.g., streptavidin-coated solid support) [17,31–33,40].

In full analogy to antibody phage display technology [41], it is good practice to quantify the efficiency of the affinity capture. The use of quantitative PCR in model screening experiments (e.g., by spiking binders into large libraries) has recently been described [42].

Solution-phase selections

Solution-phase technologies allow the identification of hits without the use of a solid-phase affinity capture step. Various methodologies can be considered. The group of David Liu [43] proposed and successfully implemented the use of interaction-dependent PCR (IDPCR) [43] for the selective PCR amplification of barcodes associated with small molecules, capable of binding to the target protein of interest. The technology relies on the use of protein–oligonucleotide conjugates, capable of annealing to (and thus acting as primer for) the single-stranded DNA structures, attached to the chemical compounds in the library. The formation of a complex between a given small molecule and the target protein of interest stabilizes the DNA heteroduplex formation and facilitates the initiation of a PCR amplification step. This methodology has recently been extended, allowing the use of protein mixtures for the simultaneous identification of cognate ligands to multiple target proteins [44]. Instead of covalently attaching oligonucleotides to the protein of interest, DNA can be coupled to an antibody or to a SNAP-tagged protein [45].

Hansen *et al.* described Binder Trap Enrichment (BTE) as a screening methodology which takes place in water-in-oil emulsions [19]. The DNA-tagged protein target and the encoded library are encapsulated in water-in-oil droplets, which create physically isolated microenvironments, in which a splint ligation step can take place between DNA-tagged protein and cognate DNA-tagged ligands only if they are bound (i.e., found in the same water droplet).

A novel method, called "DNA-programmed photoaffinity labeling" (DPAL) [46,47] features the use of small molecule–oligonucleotide conjugates, capable of forming a stable heteroduplex with a short complementary oligonucleotide, bearing a photoreactive group capable of covalent bond formation with protein residues [48]. When library members bind to the protein target in solution, an irradiation step ensures that preferential binders are "captured" on the target protein by photo-crosslinking.

Finally, members of DNA-encoded chemical libraries capable of protein binding can also be separated from nonbinding compounds using capillary electrophoresis [49] or similar flow-based technologies.

A fundamental issue in DNA-encoded chemistry relates to the largest library size that can be screened, as well as to the efficiency of the selection process. The problem can be tackled from different perspectives. On one hand, quantitative PCR methodologies provide direct experimental evidence regarding the efficiency and selectivity of ligand recovery [42,50]. Alternatively, certain estimates can be performed, using a computational analysis of selection results [51,52]. Many experimental factors may influence screening performance. In addition to ligand affinity, the practical implementation of capture methodologies [51] and washing conditions [32] can have a profound impact on experimental results (e.g., enrichment factors and reproducibility) [53].

Library decoding

After the selection step (irrespective of whether this is performed with a solid-phase capture or in solution),

18733468, 2018, 12, Downloaded from https://febs.onlinelibrary.wiley.com/doi/10.1002/1873-3468.13068 by Schweizerische Akademie Der Medizinischen Wissenschaften (Samw), Wiley Online Library on [03/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

the barcodes of preferential binders are amplified by PCR and read by high-throughput DNA sequencing. These results are deconvoluted by determining sequence counts for all library members, ideally comparing the frequency of barcodes before and after selection. We generally assume that the relative frequency, at which individually barcodes are found in the DNA sequencing experiment, corresponds to the relative abundance of the corresponding library members. The output data of library selections can be statistically evaluated [39] and represented by a suitable graphical display (Fig. 3). The relative abundance for individual library members can be visualized on two- or three-dimensional plots (called "fingerprints"). For example, when considering a library consisting of two sets of building blocks (A and B, respectively), the various A_iB_j combinations are displayed in the xy plane, while the corresponding numbers of counts (or the normalized enrichment factor) can be visualized in the z axis (not shown) or as spheres of different color and/or size (Fig. 3A). The display of selection results of a library based on three sets of building blocks requires a four-dimensional matrix. The three sets of building blocks (A, B, and C, respectively) are generally displayed in the xyz space, while enrichment factors can be represented as spheres of different color and/or size (Fig. 3B).

As shown in Fig. 3, a successful selection experiment exhibits a clear difference in sequence counts for the various library members before and after the capture step. The enriched compounds are then resynthesized, in order to confirm the experimental findings and to measure affinity constants.

Ligands isolated from DNA-encoded chemical libraries

In this section, we report on some important hits that have been discovered by different implementations of DECL technology. The first example described in Table 1 (entry a) derives from a structurally compact

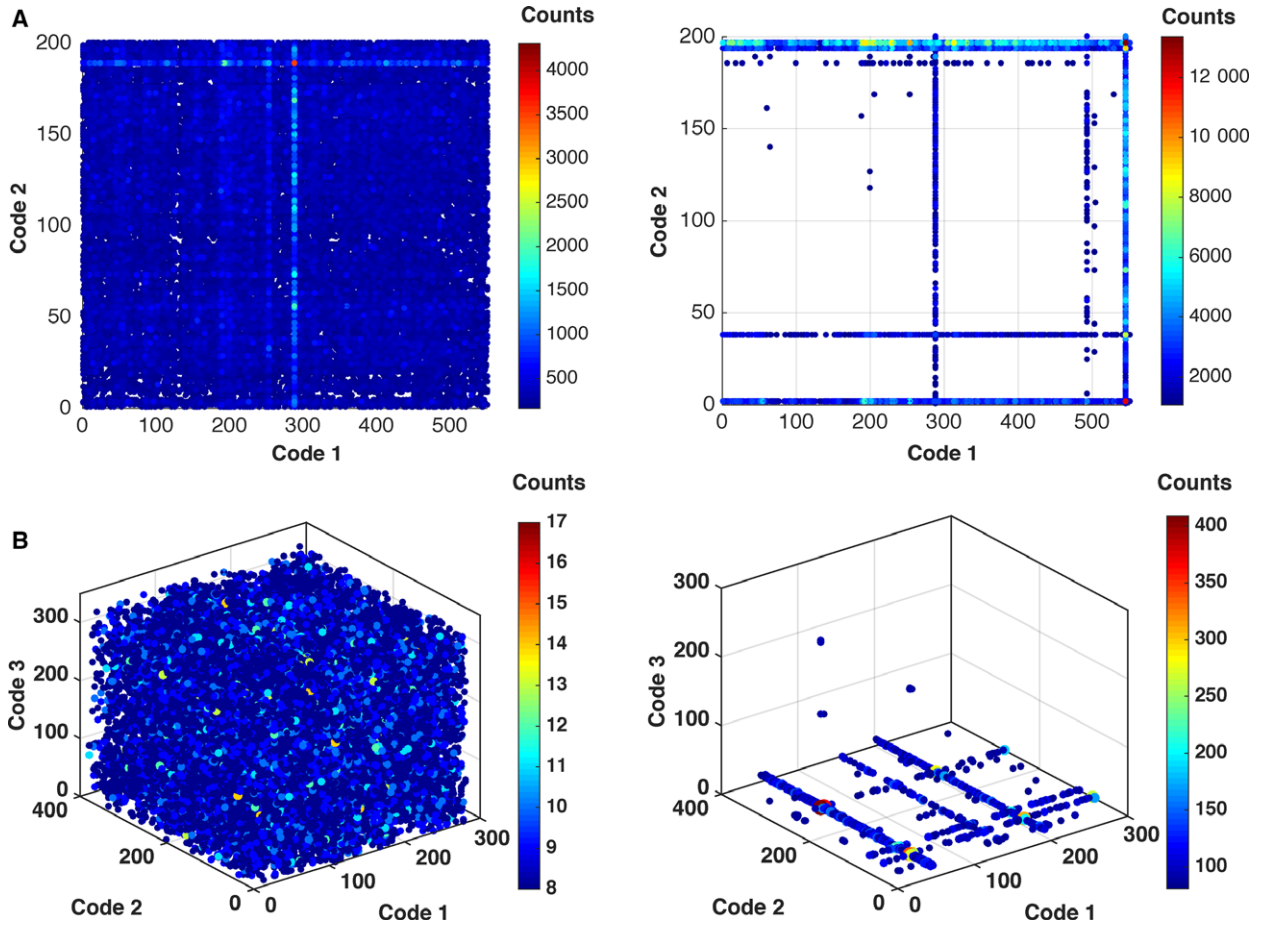
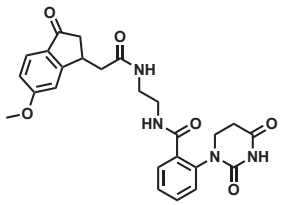
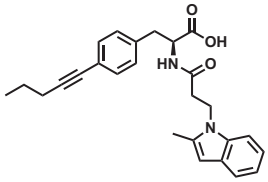
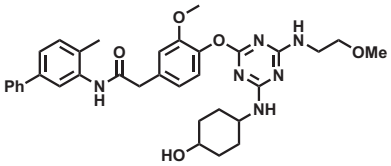
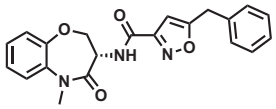
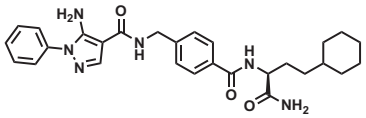
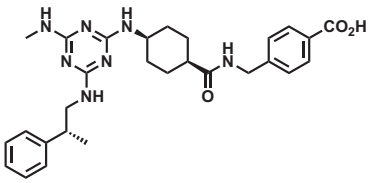
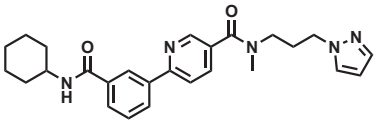


Fig. 3. Fingerprints of naïve libraries (left panels) composed of (A) two sets of building blocks or (B) three sets of building blocks are compared with the same libraries selected against (A) horseradish peroxidase (HRP) or (B) carbonic anhydrase (CA) IX (right panels).

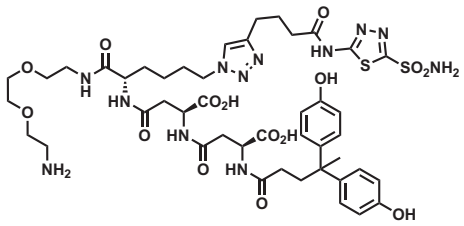
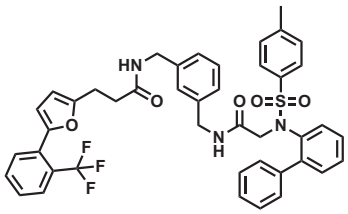
18733468, 2018, 12, Downloaded from https://febs.onlinelibrary.wiley.com/doi/10.1002/1873-3468.13068 by Schweizerische Akademie Der Medizinischen Wissenschaften (Samw), Wiley Online Library on [03/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 1. Examples of hit compounds identified from DNA-encoded chemical libraries. The following abbreviations were used for the protein targets: TNKS1, tankyrase-1; IL2, interleukin-2; NK3, Neurokinin 3; RIP1, receptor-interacting protein 1 kinase; MAPK14, mitogen-activated protein kinase 14; sEH, soluble epoxide hydrolase; CAIX, carbonic anhydrase IX; AGP, alpha-1-acid glycoprotein.

	Structure	K_d	Target	References
a)		290 nM (IC_{50})	TNKS1	[33]
b)		2.5 μM	IL2	[32]
c)		2 nM (IC_{50})	NK3	[55]
d)		10 nM (IC_{50})	RIP1	[56]
e)		7 nM (IC_{50})	MAPK14	[58]
f)		7.9 nM (IC_{50})	sEH	[50]
g)		2 nM	sEH	[59]

18733468, 2018, 12, Downloaded from https://febs.onlinelibrary.wiley.com/doi/10.1002/1873-3468.13068 by Schweizerische Akademie Der Medizinischen Wissenschaften (Samw), Wiley Online Library on [03/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 1. (Continued).

	Structure	K_d	Target	References
h)		2.6 nM	CAIX	[31]
i)		9.9 nM	AGP	[22]

two building block single-pharmacophore DNA-encoded chemical library [33]. The library contained 103‘200 members formed from two sets of carboxylic acids (240 × 430). The hit inhibits tankyrase-1 with an IC_{50} value of 290 nM, as measured by an enzymatic PARylation assay [54].

A ligand against interleukin-2 is reported in Table 1 (entry b). The library, constructed by Leimbacher *et al.* [32], was based on 100 different amino acids that had been coupled to 300 different carboxylic acids. The binder showed a dissociation constant (K_d) value of 2.5 μM, as measured by fluorescence polarization. The importance of the 2-methylindole moiety for interleukin-2 recognition was assessed by a comparative analysis of more than 100 different selections [32].

GlaxoSmithKline (GSK) isolated a 2 nM antagonist for Neurokinin 3 (NK3) [55]. The hit compound, reported in Table 1 (entry c) was isolated from a 41-million member single-pharmacophore DECL using three sets of building blocks (64 hydroxy- or amino acids, 854, and 758 amines, respectively) [55]. The building blocks were connected to DNA through a central 1,3,5-triazine scaffold. In the validated hit (Table 1, entry c), the DNA moiety had been replaced by a 2-methoxyethan-1-amino linker [55].

A first-in-class specific inhibitor of receptor-interacting protein 1 (GSK2982772) developed by GSK has entered phase 2a clinical trials for psoriasis, rheumatoid arthritis, and ulcerative colitis [56]. This compound was derived from a candidate (IC_{50} = 10 nM) (Table 1 (entry d)) that originated from a split-and-pool DECL library with three sets of building blocks, two of which proved to be essential for RIP1 binding [57].

A highly specific 7 nM inhibitor of mitogen-activated protein kinase 14 (MAPK14, also called p38-α kinase) [58] was identified by Vipergen using a yoctoReactor™ DECL. The library consisted of three sets of building blocks (two sets of amino acids and one set of carboxylic acids) and had a size of 12.6 million members [18]. Affinity selections were performed in solution using Binder Trap Enrichment (BTE) technology [19]. Table 1 (entry e) reports the hit compound of the most enriched combination in which the DNA moiety had been replaced by an NH_2 group.

Using a proprietary “Encoded Library Technology” (ELT) platform, GSK discovered an 8 nM inhibitor of soluble epoxide hydrolase (sEH) (Table 1 (entry f)), which was subsequently optimized to a candidate with IC_{50}= 27 pM (GSK2256294). This compound constitutes the first clinical candidate, which originated from DECL technology [50].

A potent inhibitor of soluble epoxide hydrolase (sEH) was also developed by X-Chem using an innovative DECL encoding strategy, featuring a DNA-ligation step that made use of Cu-catalyzed alkyne-azide cycloaddition (CuAAC) [59]. A 334-million member library was synthesized using three sets of building blocks (2259 primary amines, 222 bromoaryl acids, and 667 boronic acids). The hit compound Table 1 (entry g) in which the DNA portion was substituted by a methyl group showed a 2 nM IC_{50} for sEH.

18733468, 2018, 12, Downloaded from https://febs.onlinelibrary.wiley.com/doi/10.1002/1873-3468.13068 by Schweizerische Akademie Der Medizinischen Wissenschaften (Samw), Wiley Online Library on [03/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Dual-pharmacophore libraries based on ESAC technology [10] have also been successful [31]. A library containing more than 100 000 compounds allowed the isolation of an acetazolamide derivative (Table 1, entry h) that bound to carbonic anhydrase IX with a dissociation constant in the subnanomolar range [31]. The same ESAC library also produced a 190 nM binder against alpha-1-acid glycoprotein (AGP), thanks to the synergistic action of two building blocks that mediated a potent chelate binding to the target protein. Interestingly, the individual building blocks did not display any detectable protein binding when used on their own. The optimization of the linker connecting the two building blocks led to an improved binder Table 1 (entry i), with a dissociation constant of 9.9 nM [22].

The road ahead: challenges and opportunities

DECL technology works reliably for the isolation of hits against many target proteins of pharmaceutical interest. However, unlike human antibody technology [6], DECL sometimes fails to deliver hits to certain classes of protein targets, thus underlining the need for further technical improvements.

Thanks to advances in the quantification of library selection procedures through quantitative PCR methodologies, it is becoming increasingly clear that ligands with dissociation constants in the micromolar range are not efficiently recovered from encoded libraries in most experimental settings [60]. However, conventional technologies based on affinity capture methods are quite reliable for the isolation of nanomolar binders from encoded libraries, if and when they are present in the repertoire [42,61].

Certain classes of proteins (e.g., integral membrane proteins) are not compatible with standard selection procedures, as they are difficult to keep in solution and to capture on solid supports (e.g., streptavidin-coated beads or resin) [23]. The use of stabilized GPCRs, selection on cells or on nanodisks may be considered [55,62,63].

DNA sequencing is an essential component of DECL technology and it would be desirable to further enhance the number of DNA tags that can be sequenced. At present, one lane of Illumina™ sequencing may allow the identification of up to 200 million sequence tags. As it is possible to synthesize libraries containing billions of members, it would be important to sequence billions of tags, in order to adequately characterize selection experiments with suitable statistical tools [39].

In principle, the creativity of Chemistry is unlimited and it is likely that future success in DECL technology will derive from the practical implementation of novel DNA-compatible reactions. The set of reactions which can be considered for library synthesis is constantly expanding [64–66]. However, it is important to remember that even simple transformations (such as the formation of amide bonds) do not proceed with complete yields on 100% of substrates. An extensive characterization and selection of building blocks, which give acceptable yields, represents an essential component for any new library synthesis effort. On the other hand, preserving the integrity of DNA is important [67].

Novel scaffolds and novel concepts in library design will likely contribute to the discovery of ligands to proteins of pharmaceutical interest. We have recently observed that the central chemical moiety, used for the display of two sets of building blocks, can greatly contribute to the affinity of the resulting hit, at least for a certain proportion of target proteins [68]. Macrocyclic scaffolds may facilitate the isolation of hits against "difficult" targets [21,36,40], but these larger chemical structures may not always be compatible with pharmaceutical applications and with oral administration procedures.

It has previously been noted that the number and quality of building blocks greatly contributes to library performance [13]. On the positive side, thousands of building blocks (e.g., carboxylic acids, amines, alkynes, azides, sulphonyl chlorides, boronates) are now commercially available at affordable prices, thus facilitating library construction activities even in the academic environment. On the other hand, large pharmaceutical industries may have access to proprietary building blocks and to privileged structures. Moreover, library size (especially for repertoires consisting of two sets of building blocks) depends on the financial resources that can be devoted to purchasing oligonucleotides and building blocks. For example, a library of 100 million compounds based on two sets of building blocks will require 20 000 oligonucleotides and 20 000 building blocks, while a three-building block library with 1 billion compounds can be created using 3000 oligonucleotides and 3000 building blocks.

ESAC technology appears to be ideally suited for the discovery of fragments, which bind to adjacent sites on the target protein of interest. At present, the optimization of linkers connecting synergistic building blocks is still empirical, even though "on-DNA" procedures may facilitate this experimental task [22,25]. It remains to be seen whether linker optimization can be embedded in library construction and selection procedures. ESAC technology relies on pure components and facilitates the "affinity maturation" of existing

18733468, 2018, 12, Downloaded from https://febs.onlinelibrary.wiley.com/doi/10.1002/1873-3468.13068 by Schweizerische Akademie Der Medizinischen Wissenschaften (Samw), Wiley Online Library on [03/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

hits, which can be paired to existing libraries in a facile manner.

In summary, DECL technology has reached a good level of maturity. Both pharmaceutical companies and academic laboratories use encoded libraries for the discovery of hits. Experimental procedures have been extensively validated and are well documented in the literature. Nonetheless, we are confident that improvements in library design, chemical procedures, selections, and DNA sequencing methodologies will continue to broaden the scope of the methodology and to make it broadly accessible to the research community.

Acknowledgements

Financial support from ETH Zürich, the Swiss National Science Foundation (310030B_163479/1 Grant and CRSII2_160699/1 Sinergia Grant), the ERC Advanced Grant 670603 "Zauberkugel" and Philochem AG is gratefully acknowledged.

References

1 Goodnow RA Jr, Dumelin CE and Keefe AD (2017) DNA-encoded chemistry: enabling the deeper sampling of chemical space. *Nat Rev Drug Discov* **16**, 131–147.
2 Brenner S and Lerner RA (1992) Encoded combinatorial chemistry. *Proc Natl Acad Sci USA* **89**, 5381–5383.
3 Kang AS, Barbas CF, Janda KD, Benkovic SJ and Lerner RA (1991) Linkage of recognition and replication functions by assembling combinatorial antibody Fab libraries along phage surfaces. *Proc Natl Acad Sci USA* **88**, 4363–4366.
4 McCafferty J, Griffiths A, Winter G and Chiswell D (1990) Phage antibodies: filamentous phage displaying antibody variable domains. *Nature* **348**, 552–554.
5 Winter G and Milstein C (1991) Man-made antibodies. *Nature* **349**, 293–299.
6 Winter G, Griffiths AD, Hawkins RE and Hoogenboom HR (1994) Making antibodies by phage display technology. *Annu Rev Immunol* **12**, 433–455.
7 Roberts RW and Szostak JW (1997) RNA–peptide fusions for the *in vitro* selection of peptides and proteins. *Proc Natl Acad Sci USA* **94**, 12297–12302.
8 Ochsner UA, Green LS, Gold L and Janjic N (2014) Systematic selection of modified aptamer pairs for diagnostic sandwich assays. *Biotechniques* **56**, 125–128, 130, 132-123.
9 Lerner RA and Brenner S (2017) DNA-encoded compound libraries as open source: a powerful pathway to new drugs. *Angew Chem Int Ed Engl* **56**, 1164–1165.
10 Melkko S, Scheuermann J, Dumelin CE and Neri D (2004) Encoded self-assembling chemical libraries. *Nat Biotechnol* **22**, 568–574.
11 Gartner ZJ, Tse BN, Grubina R, Doyon JB, Snyder TM and Liu DR (2004) DNA-templated organic synthesis and selection of a library of macrocycles. *Science* **305**, 1601–1605.
12 Halpin DR and Harbury PB (2004) DNA display II. Genetic manipulation of combinatorial chemistry libraries for small-molecule evolution. *PLoS Biol* **2**, E174.
13 Franzini RM, Neri D and Scheuermann J (2014) DNA-encoded chemical libraries: advancing beyond conventional small-molecule libraries. *Acc Chem Res* **47**, 1247–1255.
14 Zimmermann G and Neri D (2016) DNA-encoded chemical libraries: foundations and applications in lead discovery. *Drug Discov Today* **21**, 1828–1834.
15 Neri D and Lerner R (2018) DNA-encoded chemical libraries: a selection system based on endowing organic compounds with amplifiable information. *Annu Rev Biochem*, https://doi.org/10.1146/annurev-biochem-062917-012550.
16 Zhu Z, Shaginian A, Grady LC, O'Keeffe T, Shi XE, Davie CP, Simpson GL, Messer JA, Evindar G, Bream RN *et al.* (2018) Design and application of a DNA-encoded macrocyclic peptide library. *ACS Chem Biol* **13**, 53–59.
17 Clark MA, Acharya RA, Arico-Muendel CC, Belyanskaya SL, Benjamin DR, Carlson NR, Centrella PA, Chiu CH, Creaser SP, Cuozzo JW *et al.* (2009) Design, synthesis and selection of DNA-encoded small-molecule libraries. *Nat Chem Biol* **5**, 647–654.
18 Hansen MH, Blakskjær P, Petersen LK, Hansen TH, Højfeldt JW, Gothelf KV and Hansen NJV (2009) A yoctoliter-scale DNA reactor for small-molecule evolution. *J Am Chem Soc* **131**, 1322–1327.
19 Blakskjaer P, Heitner T and Hansen NJ (2015) Fidelity by design: yoctoreactor and binder trap enrichment for small-molecule DNA-encoded libraries and drug discovery. *Curr Opin Chem Biol* **26**, 62–71.
20 Buller F, Mannocci L, Scheuermann J and Neri D (2010) Drug discovery with DNA-encoded chemical libraries. *Bioconjug Chem* **21**, 1571–1580.
21 Li Y, De Luca R, Cazzamalli S, Pretto F, Bajic D, Scheuermann J and Neri D (2018) Versatile protein recognition by the encoded display of multiple chemical elements on a constant macrocyclic scaffold. *Nat Chem* **10**, 441–448.
22 Bigatti M, Dal Corso A, Vanetti S, Cazzamalli S, Rieder U, Scheuermann J, Neri D and Sladojevich F (2017) Impact of a central scaffold on the binding affinity of fragment pairs isolated from DNA-encoded self-assembling chemical libraries. *ChemMedChem* **12**, 1748–1752.

18733468, 2018, 12, Downloaded from https://febs.onlinelibrary.wiley.com/doi/10.1002/1873-3468.13068 by Schweizerische Akademie Der Medizinischen Wissenschaften (Samw), Wiley Online Library on [03/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

23 Decurtins W, Wichert M, Franzini RM, Buller F, Stravs MA, Zhang Y, Neri D and Scheuermann J (2016) Automated screening for small organic ligands using DNA-encoded chemical libraries. *Nat Protoc* **11**, 764–780.
24 Lin W, Reddavide FV, Uzunova V, Gur FN and Zhang Y (2015) Characterization of DNA-conjugated compounds using a regenerable chip. *Anal Chem* **87**, 864–868.
25 Zimmermann G, Li Y, Rieder U, Mattarella M, Neri D and Scheuermann J (2017) Hit-validation methodologies for ligands isolated from DNA-encoded chemical libraries. *ChemBioChem* **18**, 853–857.
26 Svensen N, Diaz-Mochon JJ and Bradley M (2011) Decoding a PNA encoded peptide library by PCR: the discovery of new cell surface receptor ligands. *Chem Biol* **18**, 1284–1289.
27 Zambaldo C, Barluenga S and Winssinger N (2015) PNA-encoded chemical libraries. *Curr Opin Chem Biol* **26**, 8–15.
28 Gorska K, Huang KT, Chaloin O and Winssinger N (2009) DNA-templated homo- and heterodimerization of peptide nucleic acid encoded oligosaccharides that mimick the carbohydrate epitope of HIV. *Angew Chem Int Ed Engl* **48**, 7695–7700.
29 Mannocci L, Zhang Y, Scheuermann J, Leimbacher M, De Bellis G, Rizzi E, Dumelin C, Melkko S and Neri D (2008) High-throughput sequencing allows the identification of binding molecules isolated from DNA-encoded chemical libraries. *Proc Natl Acad Sci USA* **105**, 17670–17675.
30 Liu R, Li X and Lam KS (2017) Combinatorial chemistry in drug discovery. *Curr Opin Chem Biol* **38**, 117–126.
31 Wichert M, Krall N, Decurtins W, Franzini RM, Pretto F, Schneider P, Neri D and Scheuermann J (2015) Dual-display of small molecules enables the discovery of ligand pairs and facilitates affinity maturation. *Nat Chem* **7**, 241–249.
32 Leimbacher M, Zhang Y, Mannocci L, Stravs M, Geppert T, Scheuermann J, Schneider G and Neri D (2012) Discovery of small-molecule interleukin-2 inhibitors from a DNA-encoded chemical library. *Chemistry* **18**, 7729–7737.
33 Franzini RM, Ekblad T, Zhong N, Wichert M, Decurtins W, Nauer A, Zimmermann M, Samain F, Scheuermann J, Brown PJ *et al.* (2015) Identification of structure-activity relationships from screening a structurally compact DNA-encoded chemical library. *Angew Chem Int Ed Engl* **54**, 3927–3931.
34 Kanan MW, Rozenman MM, Sakurai K, Snyder TM and Liu DR (2004) Reaction discovery enabled by DNA-templated synthesis and *in vitro* selection. *Nature* **431**, 545–549.
35 Li X and Liu DR (2004) DNA-templated organic synthesis: nature's strategy for controlling chemical reactivity applied to synthetic molecules. *Angew Chem Int Ed Engl* **43**, 4848–4870.
36 Kleiner RE, Dumelin CE, Tiu GC, Sakurai K and Liu DR (2010) *In vitro* selection of a DNA-templated small-molecule library reveals a class of macrocyclic kinase inhibitors. *J Am Chem Soc* **132**, 11779–11791.
37 Li Y, Zhao P, Zhang M, Zhao X and Li X (2013) Multistep DNA-templated synthesis using a universal template. *J Am Chem Soc* **135**, 17727–17730.
38 Chan AI, McGregor LM and Liu DR (2015) Novel selection methods for DNA-encoded chemical libraries. *Curr Opin Chem Biol* **26**, 55–61.
39 Buller F, Steiner M, Scheuermann J, Mannocci L, Nissen I, Kohler M, Beisel C and Neri D (2010) High-throughput sequencing for the identification of binding molecules from DNA-encoded chemical libraries. *Bioorg Med Chem Lett* **20**, 4188–4192.
40 Maianti JP, McFedries A, Foda ZH, Kleiner RE, Du XQ, Leissring MA, Tang WJ, Charron MJ, Seeliger MA, Saghatelian A *et al.* (2014) Anti-diabetic activity of insulin-degrading enzyme inhibitors mediated by multiple hormones. *Nature* **511**, 94–98.
41 Mutuberria R, Hoogenboom HR, Linden EVD, Bruine APD and Roovers RC (1999) Model systems to study the parameters determining the success of phage antibody selections on complex antigens. *J Immunol Methods* **231**, 65–81.
42 Li Y, Zimmermann G, Scheuermann J and Neri D (2017) Quantitative PCR is a valuable tool to monitor the performance of DNA-encoded chemical library selections. *ChemBioChem* **18**, 848–852.
43 McGregor LM, Gorin DJ, Dumelin CE and Liu DR (2010) Interaction-dependent PCR: identification of ligand-target pairs from libraries of ligands and libraries of targets in a single solution-phase experiment. *J Am Chem Soc* **132**, 15522–15524.
44 McGregor LM, Jain T and Liu DR (2014) Identification of ligand-target pairs from combined libraries of small molecules and unpurified protein targets in cell lysates. *J Am Chem Soc* **136**, 3264–3270.
45 Keppler A, Gendreizig S, Gronemeyer T, Pick H, Vogel H and Johnsson K (2003) A general method for the covalent labeling of fusion proteins with small molecules *in vivo*. *Nat Biotechnol* **21**, 86–89.
46 Li G, Liu Y, Liu Y, Chen L, Wu S, Liu Y and Li X (2013) Photoaffinity labeling of small-molecule-binding proteins by DNA-templated chemistry. *Angew Chem Int Ed Engl* **52**, 9544–9549.
47 Wang D-Y, Cao Y, Zheng L-Y, Chen L-D, Chen X-F, Hong Z-Y, Zhu Z-Y, Li X and Chai Y-F (2017) Target identification of kinase inhibitor Alisertib (MLN8237) by using DNA-programmed affinity labeling. *Chemistry* **23**, 10906–10914.
48 Zhao P, Chen Z, Li Y, Sun D, Gao Y, Huang Y and Li X (2014) Selection of DNA-encoded small molecule

18733468, 2018, 12, Downloaded from https://febs.onlinelibrary.wiley.com/doi/10.1002/1873-3468.13068 by Schweizerische Akademie Der Medizinischen Wissenschaften (Samw), Wiley Online Library on [03/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

libraries against unmodified and non-immobilized protein targets. *Angew Chem Int Ed Engl* **53**, 10056–10059.

49 Bao J, Krylova SM, Cherney LT, Hale RL, Belyanskaya SL, Chiu CH, Shaginian A, Arico-Muendel CC and Krylov SN (2016) Predicting electrophoretic mobility of protein-ligand complexes for ligands from DNA-encoded libraries of small molecules. *Anal Chem* **88**, 5498–5506.

50 Belyanskaya SL, Ding Y, Callahan JF, Lazaar AL and Israel DI (2017) Discovering drugs with DNA-encoded library technology: from concept to clinic with an inhibitor of soluble epoxide hydrolase. *ChemBioChem* **18**, 837–842.

51 Satz AL (2015) DNA encoded library selections and insights provided by computational simulations. *ACS Chem Biol* **10**, 2237–2245.

52 Satz AL (2016) Simulated screens of DNA encoded libraries: the potential influence of chemical synthesis fidelity on interpretation of structure-activity relationships. *ACS Comb Sci* **18**, 415–424.

53 Kuai L, O'Keeffe T and Arico-Muendel C (2018) Randomness in DNA encoded library selection data can be modeled for more reliable enrichment calculation. *SLAS Discov*, https://doi.org/10.1177/2472555218757718.

54 Wahlberg E, Karlberg T, Kouznetsova E, Markova N, Macchiarulo A, Thorsell AG, Pol E, Frostell A, Ekblad T, Oncu D *et al.* (2012) Family-wide chemical profiling and structural analysis of PARP and tankyrase inhibitors. *Nat Biotechnol* **30**, 283–288.

55 Wu Z, Graybill TL, Zeng X, Platchek M, Zhang J, Bodmer VQ, Wisnoski DD, Deng J, Coppo FT, Yao G *et al.* (2015) Cell-based selection expands the utility of DNA-encoded small-molecule library technology to cell surface drug targets: identification of novel antagonists of the NK3 Tachykinin receptor. *ACS Comb Sci* **17**, 722–731.

56 Harris PA, Berger SB, Jeong JU, Nagilla R, Bandyopadhyay D, Campobasso N, Capriotti CA, Cox JA, Dare L, Dong X *et al.* (2017) Discovery of a first-in-class receptor interacting protein 1 (RIP1) kinase specific clinical candidate (GSK2982772) for the treatment of inflammatory diseases. *J Med Chem* **60**, 1247–1261.

57 Harris PA, King BW, Bandyopadhyay D, Berger SB, Campobasso N, Capriotti CA, Cox JA, Dare L, Dong X, Finger JN *et al.* (2016) DNA-encoded library screening identifies Benzo[b][1,4]oxazepin-4-ones as highly potent and monoselective receptor interacting protein 1 kinase inhibitors. *J Med Chem* **59**, 2163–2178.

58 Petersen LK, Blakskjær P, Chaikuad A, Christensen AB, Dietvorst J, Holmkvist J, Knapp S, Kořínek M, Larsen LK, Pedersen AE *et al.* (2016) Novel p38α MAP kinase inhibitors identified from yoctoReactor DNA-encoded small molecule library. *Med Chem Commun* **7**, 1332–1339.

59 Litovchick A, Dumelin CE, Habeshian S, Gikunju D, Guie MA, Centrella P, Zhang Y, Sigel EA, Cuozzo JW, Keefe AD *et al.* (2015) Encoded library synthesis using chemical ligation and the discovery of sEH Inhibitors from a 334-million member library. *Sci Rep* **5**, 10916.

60 Denton KE, Wang S, Gignac MC, Milosevich N, Hof F, Dykhuizen EC and Krusemark CJ (2017) Robustness of *in vitro* selection assays of DNA-encoded peptidomimetic ligands to CBX7 and CBX8. *SLAS Discov*, https://doi.org/10.1177/2472555217750871.

61 Cuozzo JW, Centrella PA, Gikunju D, Habeshian S, Hupp CD, Keefe AD, Sigel EA, Soutter HH, Thomson HA, Zhang Y *et al.* (2017) Discovery of a potent BTK inhibitor with a novel binding mode by using parallel selections with a DNA-encoded chemical library. *ChemBioChem* **18**, 864–871.

62 Bayburt TH and Sligar SG (2010) Membrane protein assembly into Nanodiscs. *FEBS Lett* **584**, 1721–1727.

63 Klenk C, Ehrenmann J, Schutz M and Pluckthun A (2016) A generic selection system for improved expression and thermostability of G protein-coupled receptors by directed evolution. *Sci Rep* **6**, 21294.

64 Satz AL, Cai J, Chen Y, Goodnow R, Gruber F, Kowalczyk A, Petersen A, Naderi-Oboodi G, Orzechowski L and Strebel Q (2015) DNA compatible multistep synthesis and applications to DNA encoded libraries. *Bioconjug Chem* **26**, 1623–1632.

65 Skopic MK, Salamon H, Bugain O, Jung K, Gohla A, Doetsch LJ, Dos Santos D, Bhat A, Wagner B and Brunschweiger A (2017) Acid- and Au(i)-mediated synthesis of hexathymidine-DNA-heterocycle chimeras, an efficient entry to DNA-encoded libraries inspired by drug structures. *Chem Sci* **8**, 3356–3361.

66 Skopic MK, Willems S, Wagner B, Schieven J, Krause N and Brunschweiger A (2017) Exploration of a Au(I)-mediated three-component reaction for the synthesis of DNA-tagged highly substituted spiroheterocycles. *Org Biomol Chem* **15**, 8648–8654.

67 Malone ML and Paegel BM (2016) What is a "DNA-compatible" reaction? *ACS Comb Sci* **18**, 182–187.

68 Favalli N, Biendl S, Hartmann M, Piazzi J, Sladojevich F, Gräslund S, Brown PJ, Näreoja K, Schüler H, Scheuermann J *et al.* (2018) A DNA-encoded library of chemical compounds based on common scaffolding structures reveals the impact of ligand geometry on protein recognition. in press.

18733468, 2018, 12, Downloaded from https://febs.onlinelibrary.wiley.com/doi/10.1002/1873-3468.13068 by Schweizerische Akademie Der Medizinischen Wissenschaften (Samw), Wiley Online Library on [03/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License